

Physics of the Lung and Breathing

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Objectives

- Explain the physics involved in the exchange of gas between the lungs and the blood.,
- The breathing mechanism.
- Interaction between the cardiovascular and pulmonary system.
- Effect of surface tension of breathing mechanism and lung function.
- Definition to Pressure- Volume Curve. Normal Alveoli, collapsed alveoli.

Objectives

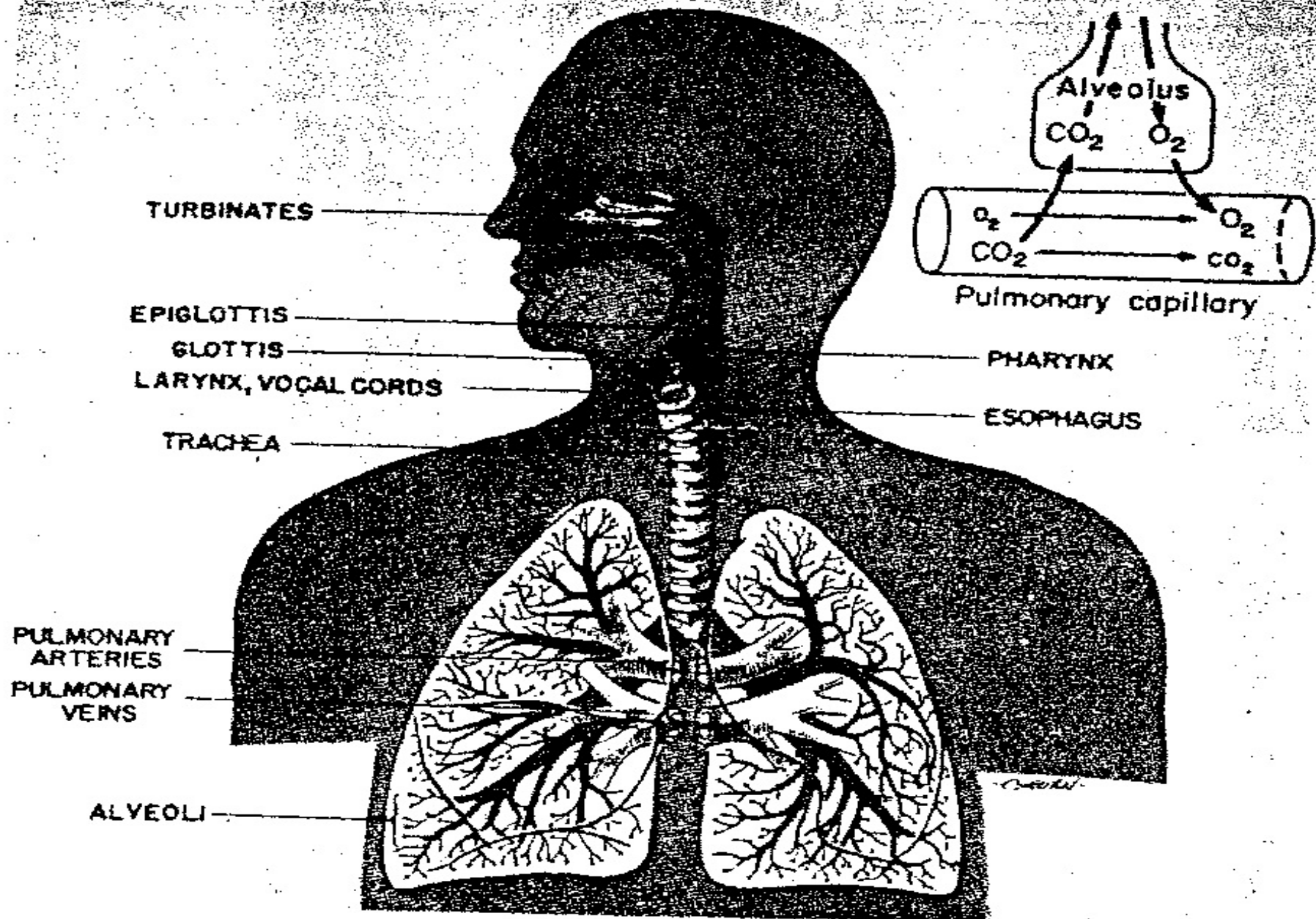
- Compliance normal range, Low Compliance, and Large Compliance.
- Perfusion, ventilation, and Diffusion
- How lung volumes measured, Spirometer.
- The physical aspects of some common lung diseases.

Introduction

- The body is in ways a very remarkable machine
- combined with O_2 in the cells to release energy
- Its by-products are disposed
 - the non-digestible solid
 - water and some other
 - almost 0.5 kg of CO_2 is disposed
 - heat is dissipated from the body's surface.

Introduction

- The human consists of trillion living cells
- The lungs (pulmonary system) serve as the supplier of O_2 and the disposer of the main by-product, CO_2 .
- The blood and its vessels (cardiovascular system) serve as the transport system
 - Each of these miniature engines must be provided with fuel, O_2 ,
 - and a method of getting rid of the by-products



Introduction

- ❑ Other Functions of the Lungs
 - Exchange of O_2 and CO_2
 - Keeps the bodies pH (acid) constant
 - Moisturizes the air
 - Play a role in heat exchange
 - Voice production (power of 1mW)
 - Air for yawning sighing, laughing, sniffing

Breathing

- We breath about 6 liters of air per minute
 - This is about the same as the heart pumps
- Men breath about 15/minute
- Women breath about 18/minute
- Infants' breath abut 60/minute

What We Breath In and Out

- The air we inspire is about 80% N_2 and 20% O_2
- Expired air is about 80% N_2 , 16% O_2 , and 4% CO_2 .
- This relatively high percentage of O_2 of in exhaled air mouth-to-mouth resuscitation

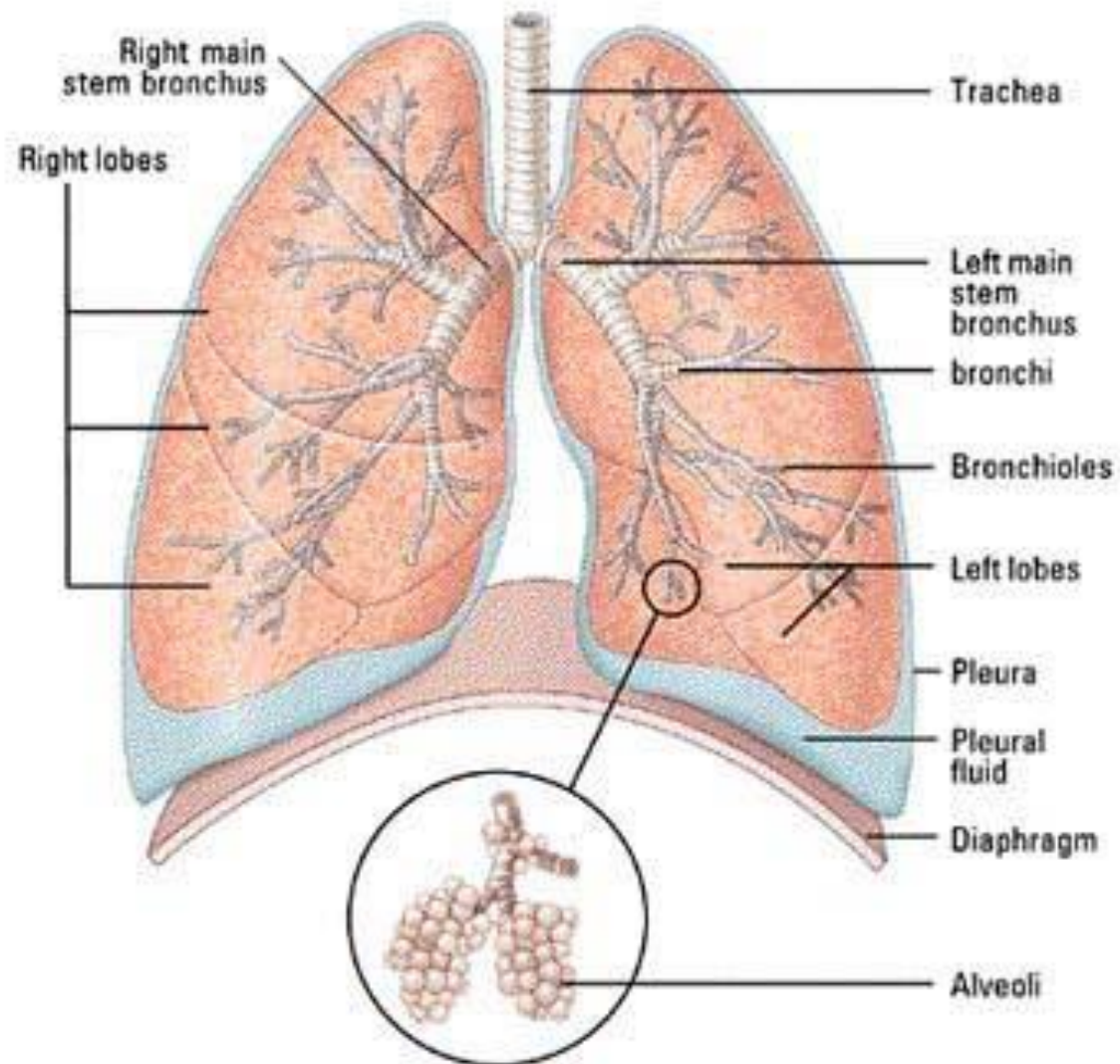
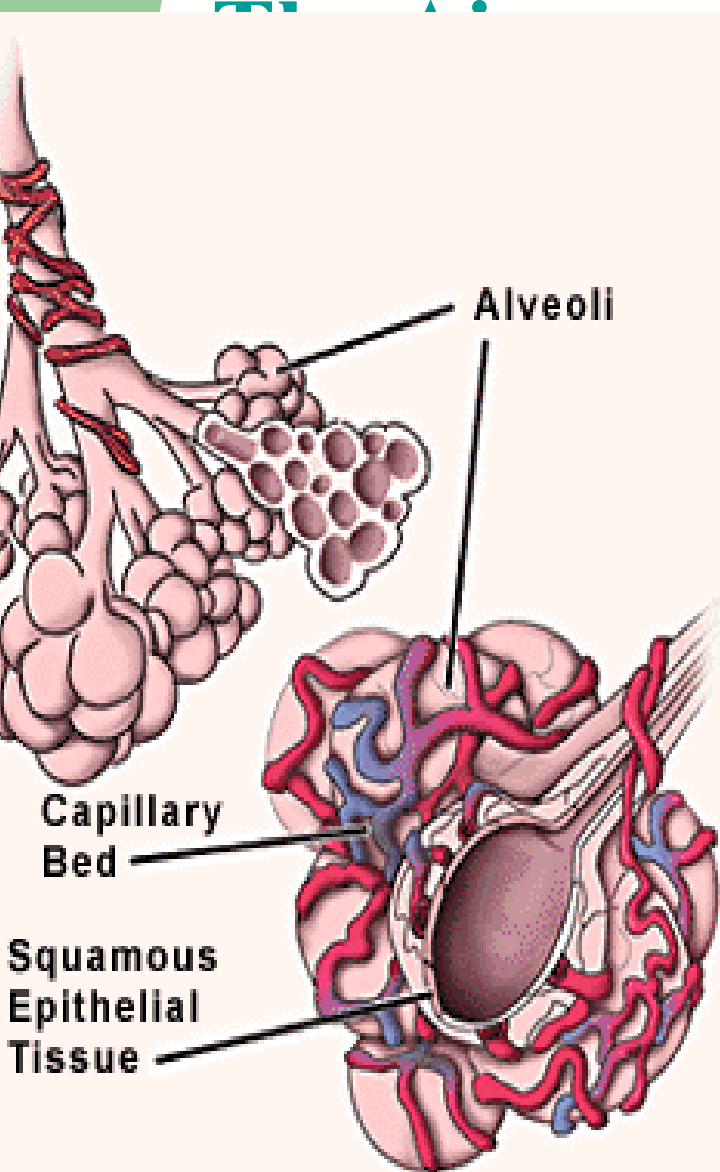
What We Breath In and Out

- We breathe about 10 kg of air each day
- The lungs absorb about 0.5 kg of O_2 ;
 - release a slightly smaller amount of CO_2 .
 - saturate the air we breathe with water.
- When we breath in dry air,
 - Expired air carries away about 0.5 kg of water each day

The Airways

- ❑ The alveoli, which are like small interconnected bubbles shown in the Fig below,
 - About 0.2mm in diameter
 - have walls only 0.4 μ m thick.
 - They expand and contract during breathing
 - are “where the action is” in the exchange of O₂ and CO₂.
- ❑ Lung has a surface area of about 80m²

Lungs



The Airways

- ❑ Each alveolus is surrounded by blood so that
 - O_2 can diffuse from the alveolus into the red blood cells
 - CO_2 can diffuse from the blood into the air in the alveolus.

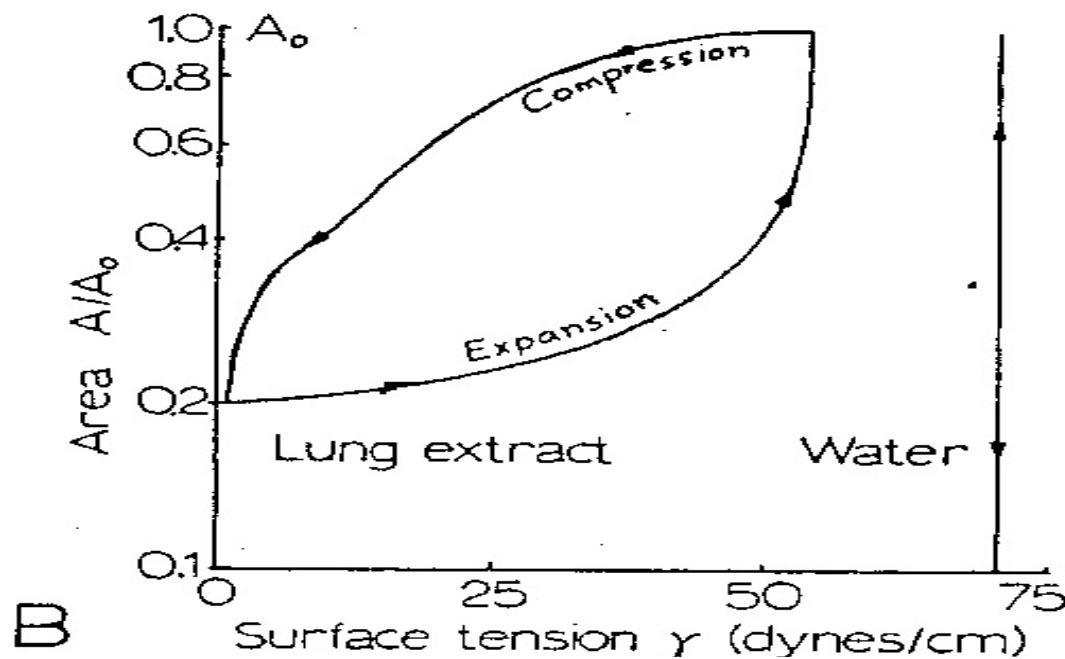
The Airways

- ❑ At birth the lungs have about 30 million alveoli;
 - by age 8 the number of alveoli has increased to about 300 million.
 - Beyond this age the number stays relatively constant but the alveoli increase in diameter.

Physics of the Alveoli

- **Alveoli** have a natural tendency to get smaller due to the surface tension of their unique fluid lining.
- This lining, a type of surfactant, is necessary for the lung to function properly.
- The absence of surfactant in the lungs of some newborn infants,
 - idiopathic respiratory distress syndrome (RDS), hyaline membrane disease

Physics of the Alveoli



The surface tension of a film of normal lung extract containing surfactant.

Physics of the Alveoli

- The surface tension of the surfactant that lines the alveoli of healthy individuals plays a major role in lung function.
- The surface tension of the surfactant is not constant. $p=4\gamma /R$ a form of Laplace law.
- large decrease of γ as the area decreases

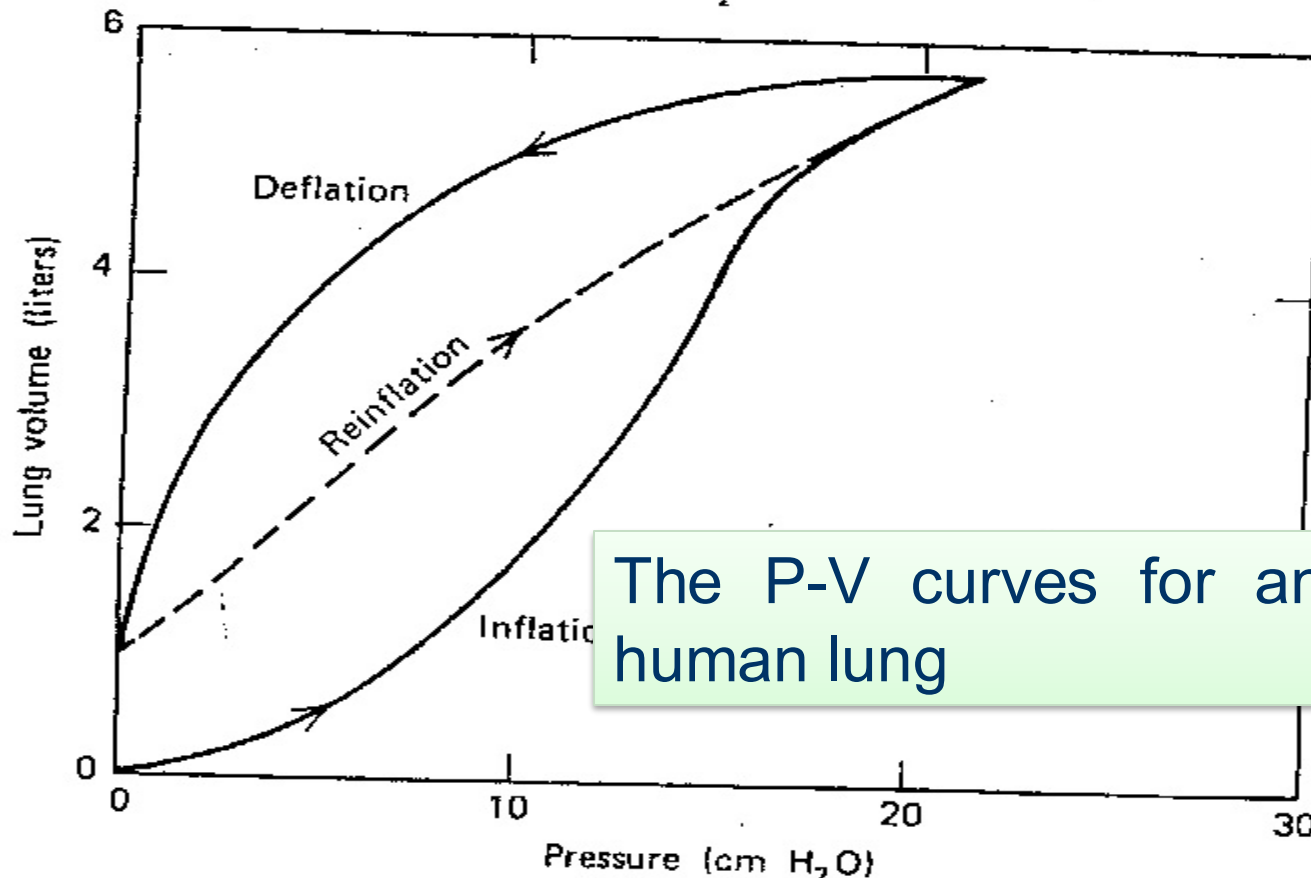
Physics of the Alveoli

- this causes the alveolus to stabilize at about one-fourth its maximum size.
- Alveoli not covered with surfactant —→, RDS,
 - collapse like small bubbles, and quite a large pressure is needed to reopen them.

Physics of the Alveoli

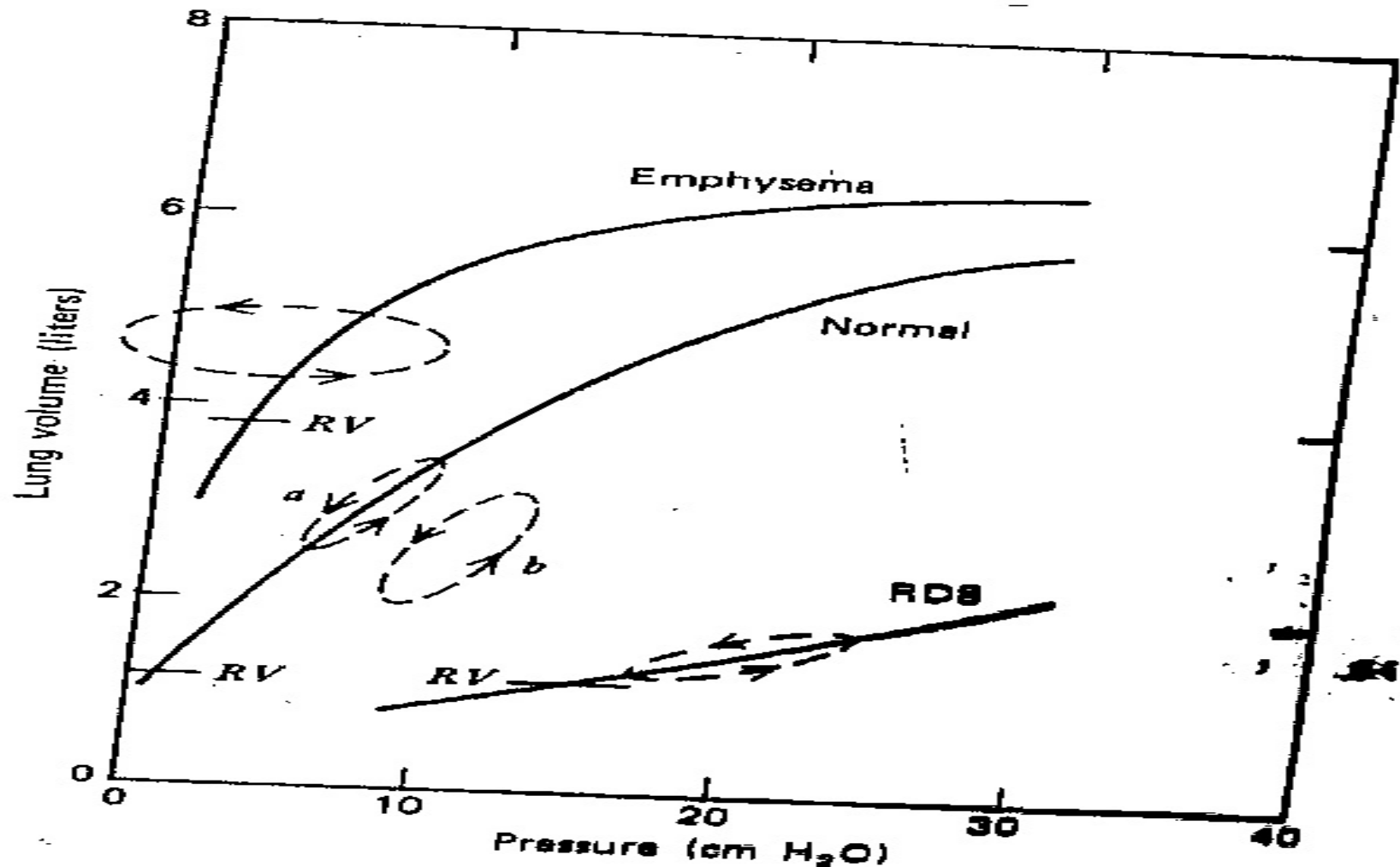
- An infant with RDS may not have the energy to breath with its low compliance lungs.
- Therapy may involve positive pressure breathing to help open the alveoli.

Physics of the Alveoli



The P-V curves for an excised human lung

P-V curve



Pressure Volume relationship

- During normal tidal breathing the hysteresis loop is quite small.
- If tidal breathing continues unchanged,
 - some of the alveoli collapse,
 - the hysteresis loop becomes slightly larger and shifts toward higher pressure.

Pressure Volume relationship

- a deep breath reopens the alveoli,
 - and the curve shifts back to **a**.
 - deep breath occasionally without being aware of it- a sigh.
- ❑ During surgery the anesthesiologist will occasionally force a large volume of gas into the patient's lungs to reopen collapsed alveoli.

Pressure Volume relation ship

- ❑ Sever emphysema ;the compliance $c = (\Delta V / \Delta P)$;
 - the increased compliance,
 - the larger residual volume,
 - the large area inside the hysteresis loop.
- ❑ The P-V curve for an infant lung suffering from RDS
 - The hysteresis loop is shifted to the right
 - A large pressure must be maintained to keep the lung inflated.
 - The lower compliance $(\Delta V / \Delta P)$ of this lung

How the Blood and Lungs Interact.

- The average peak blood pressure in the main pulmonary artery = 20 mm Hg
 - About one fifth (liter) of the body's blood supply is in the lungs,
 - only about 70ml of that blood is in the capillaries of the lungs at any one time.

How the Blood and Lungs Interact

- Since blood is in the pulmonary capillaries for less than 1S,
- the lungs have extremely thin walls and are surrounded by the blood in the pulmonary capillary system.
- The surface area between air and blood in the lungs is about 80 m².
 - If the 70ml of blood in the pulmonary capillaries were spread over a surface area of 80m²,

How the Blood and Lungs Interact

- the resulting layer of blood would be only about $1\mu\text{m}$ thick
- Two general processes are involved in gas exchange in the lungs:
 - (*perfusion*).
 - (*ventilation*).
- If either process fails not all the blood will be properly oxygenated.

How the Blood and Lungs Interact

- ❑ There are three types of ventilation-perfusion areas in the lungs:
 - good ventilation and good perfusion,
 - good ventilation and poor perfusion
 - poor ventilation and good perfusion

The transfer of O_2 and CO_2 in out of the blood

- diffusion in liquids.

$$D = \lambda \sqrt{N}$$

N collisions

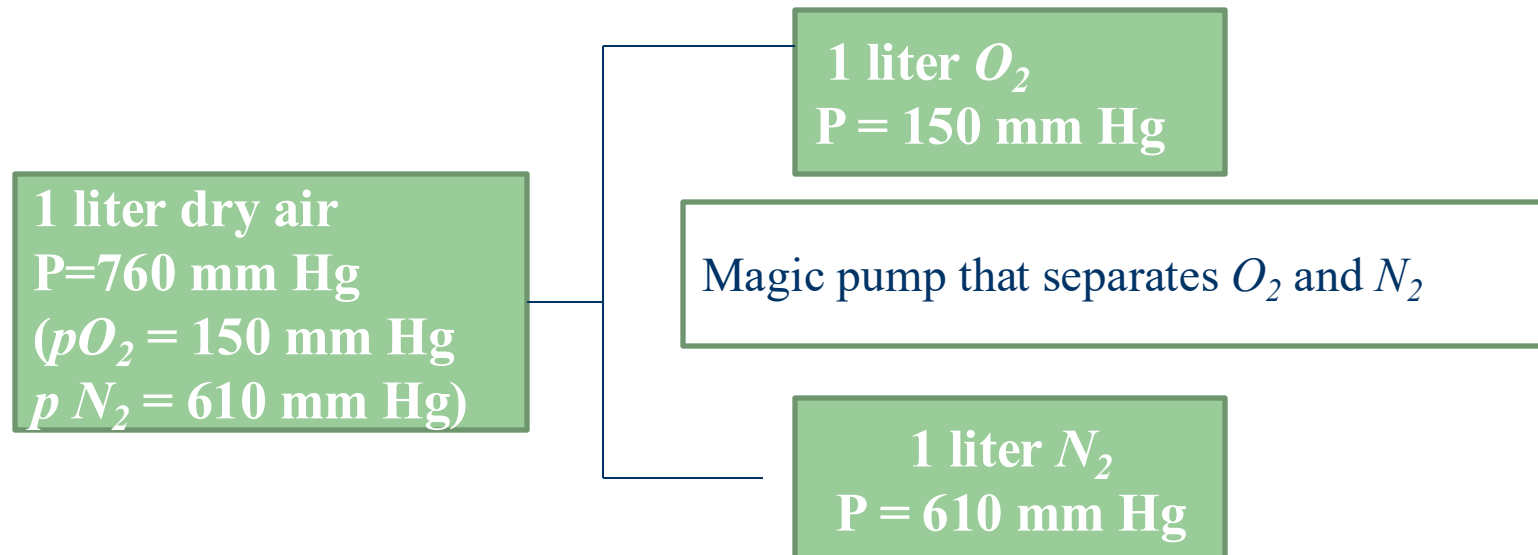
D probable distance

λ is the mean free path = 10^{-7} m in air
= 10^{-11} m in tissue

- Diffusion depends on the **speed of the molecules** .

The transfer of O_2 and CO_2 in out of the blood

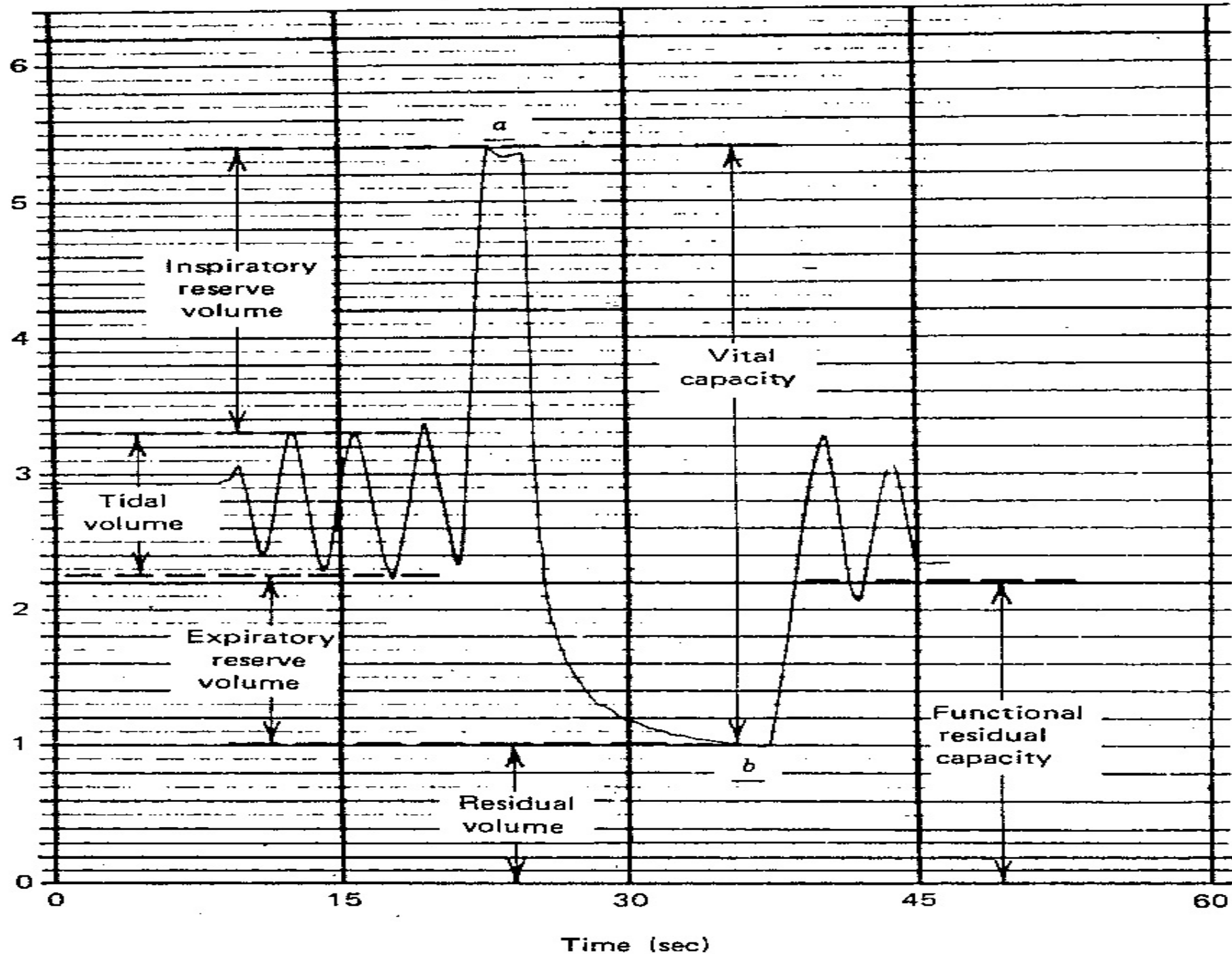
□ Dalton's law of partial pressures.



□ The partial pressure of water vapor in the lungs at **37 °C** and **100% relative humidity** = (47 mmHg)

The transfer of O_2 and CO_2 into and out of the blood

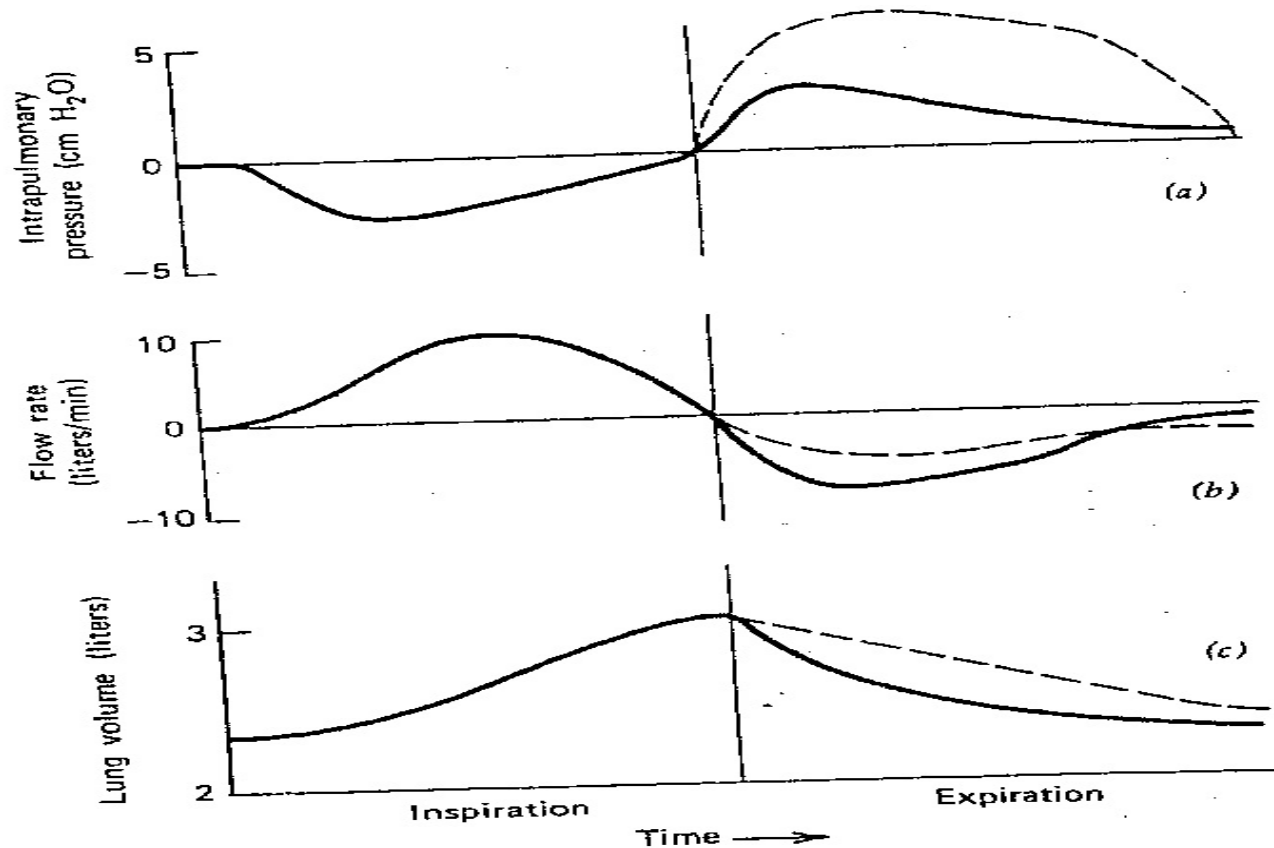
- ❑ **Henry's law of solubility of gases**
- ❑ The amount of gas dissolved in blood varies greatly from one gas to another
 - At a pCO_2 of only 40mm Hg hold about 25 cm^3 of CO_2 in each liter
 - If a disease causes the alveolar wall to thicken, the transport of O_2 is hindered more than the transport of CO_2 .
 - N_2 plays a safety role



Measurement of Lung Volumes

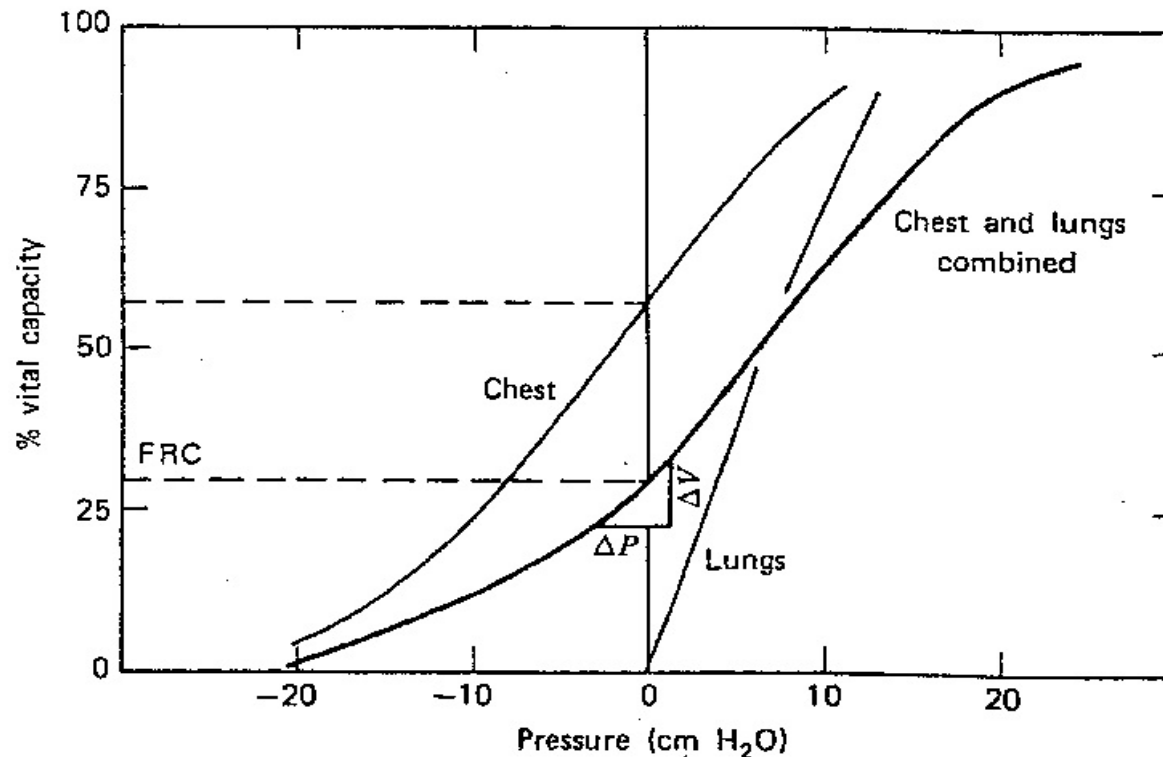
- ❑ During normal breathing at rest we inhale about 0.5 liter (500 cm^3) of air with each breath.
- ❑ ***the tidal volume at rest.***
- ❑ Two general categories of respiratory dysfunction yield abnormal results during spirometry
 - **-obstructive** lung disease
 - **restrictive** lung disease

Pressure-Airflow-Volume Relationships of the Lungs



... illustrates (b) and lung volumes (c) d

Pressure-Airflow-Volume Relationships of the Lungs



Pressure-Airflow-Volume Relationships of the Lungs

- **Compliance** is the ratio of the change in volume produced by a small change in pressure.

$$\Delta V / \Delta P \quad \text{liter/cm H}_2\text{O}$$

- normal adults is in the range of 0.18 to 0.27 liter/cm H₂O.
- 25% greater in men over age 60 than in younger men.
- There is little change in women with age.

Pressure-Airflow-Volume Relationships of the Lungs

- A stiff (fibrotic) lung
 - it has a **low compliance**.
- A flappy lung
 - has a **large compliance**
- emphysema, the compliance increases.
- Infants with RDS have lungs with low compliance.

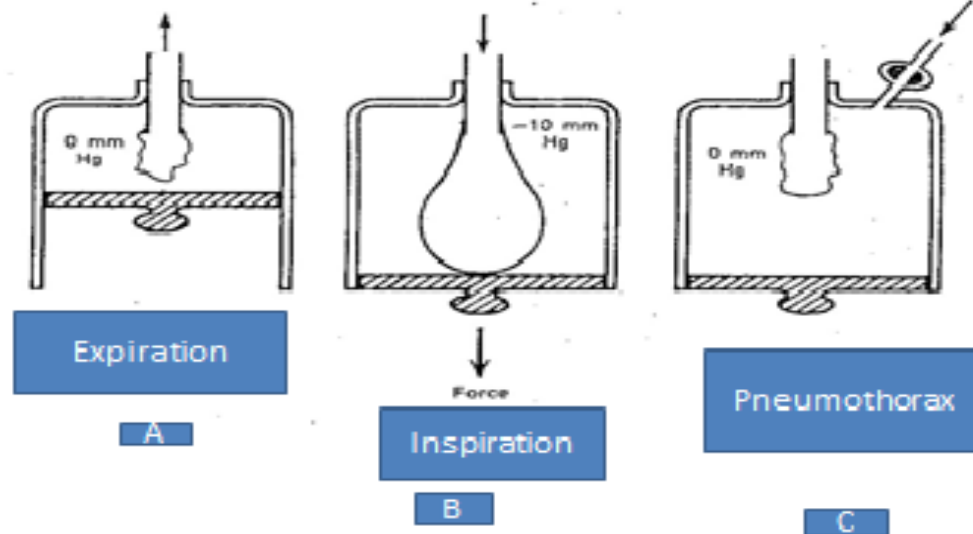
The Breathing Mechanism

- ❑ The lungs do not normally collapse because they are in an airtight container-the chest.
- ❑ As the diaphragm and rib cage move, the lungs stay in contact with them.
- ❑ Two forces keep the lungs from collapsing:
 - (1) Surface tension between the lungs and chest wall
 - (2) Air pressure inside the lungs.

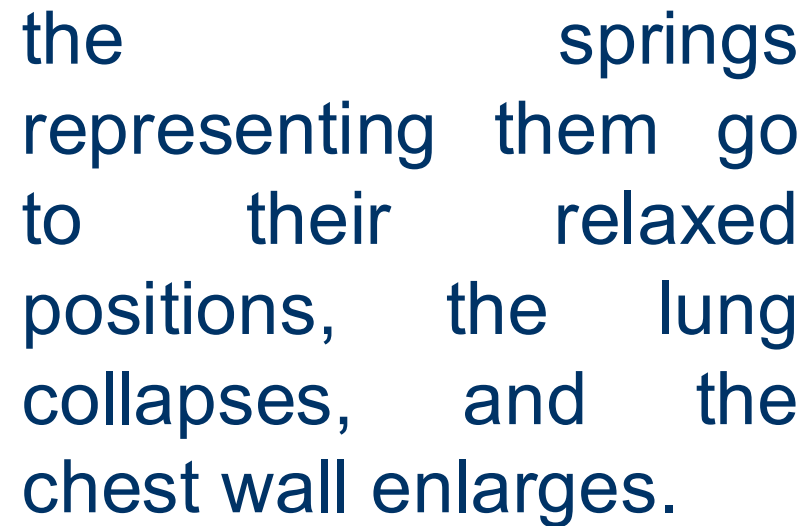
The Breathing Mechanism

- ❑ The air inside the lungs is at atmospheric pressure-100kPa
 - it would push the lungs back in contact with the chest wall.
 - There is normally a negative pressure of 6 to 12 kPa (5 to 10 mm Hg) in the intrapleural space
- ❑ Treatment of **tuberculosis**, it is medically desirable to collapse one lung to allow it to “rest”

The Breathing Mechanism



pneumothorax (literally, air-chest)



Airway Resistance

- ❑ The flow of air in the lungs is analogous to the flow of current in an electrical circuit.
- ❑ “Ohms law” for airflow looks like Ohm’s law for electrical circuits, with voltage replaced by pressure difference $\Delta V/\Delta t$.
- ❑ Airway resistance R_g depends
 - dimensions of the tube and
 - viscosity of the gas

Airway Resistance

- Most of the resistance is in the upper airway passages.
 - The nasal area accounts for about half of R_g
- another 20% is due to the other upper airways
 - normal subjects less than 10% of R_g is in the terminal airways resistance.
 - diseases that affect the terminal airways (bronchioles and alveoli) do not appreciably affect the airway resistance until they are far advanced.

Airway Resistance

- $R_g C$ = Time Constant for the lung

R_g airway resistance

C Compliance

compliance = $\Delta V / \Delta P$

Example:

If $R_g = 300 \text{ Pa(liter/s)}$, what air volume flow rate, V/s , would occur at an expiratory pressure of 13 kPa ?

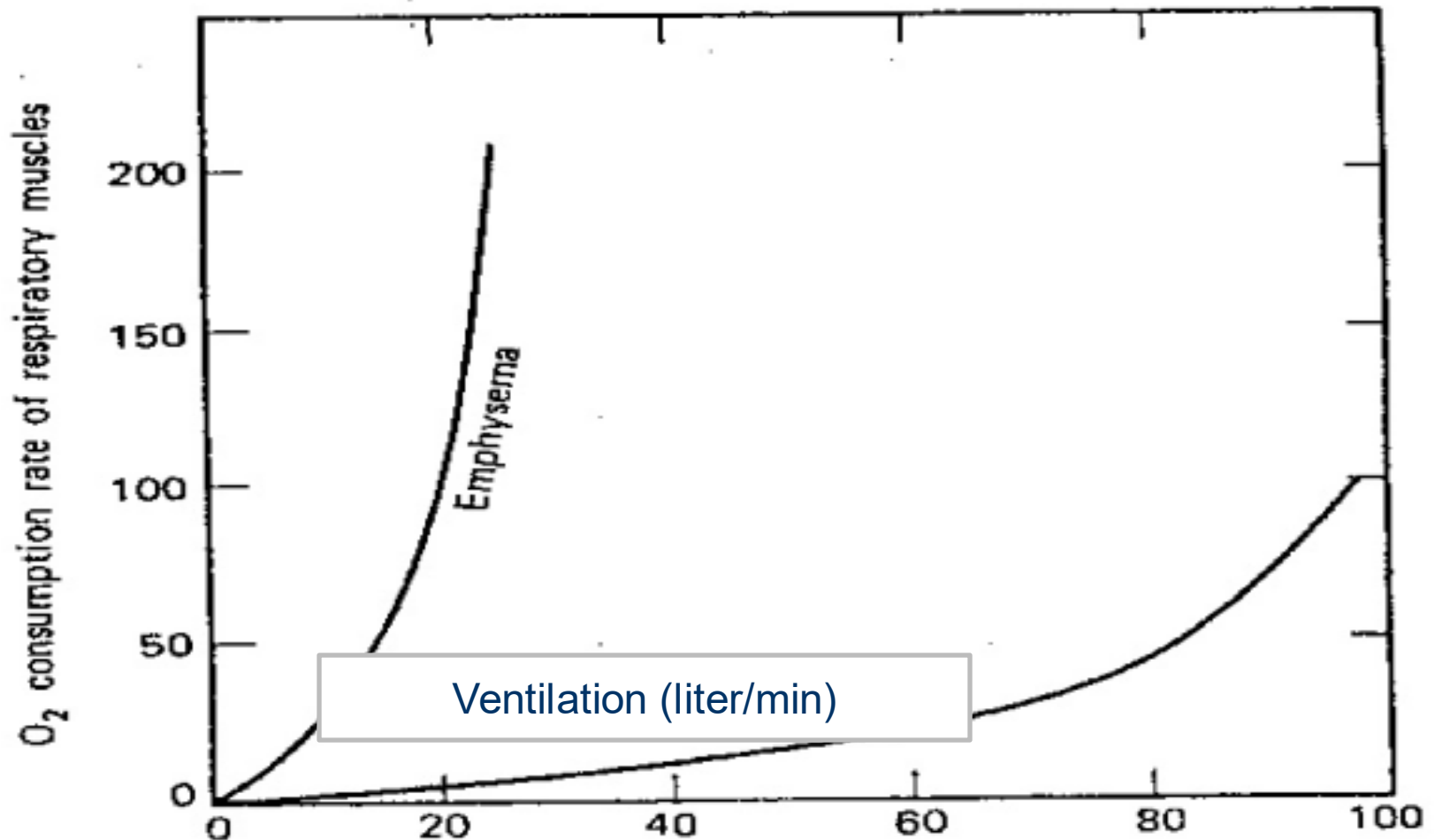
Physics of Some Common Lung Diseases

- The physics of RDS
- In emphysema
 - the divisions between the alveoli break down
 - producing larger lung spaces.
 - destruction of lung tissue reduces the springiness of the lungs.
 - The lungs become more compliant-

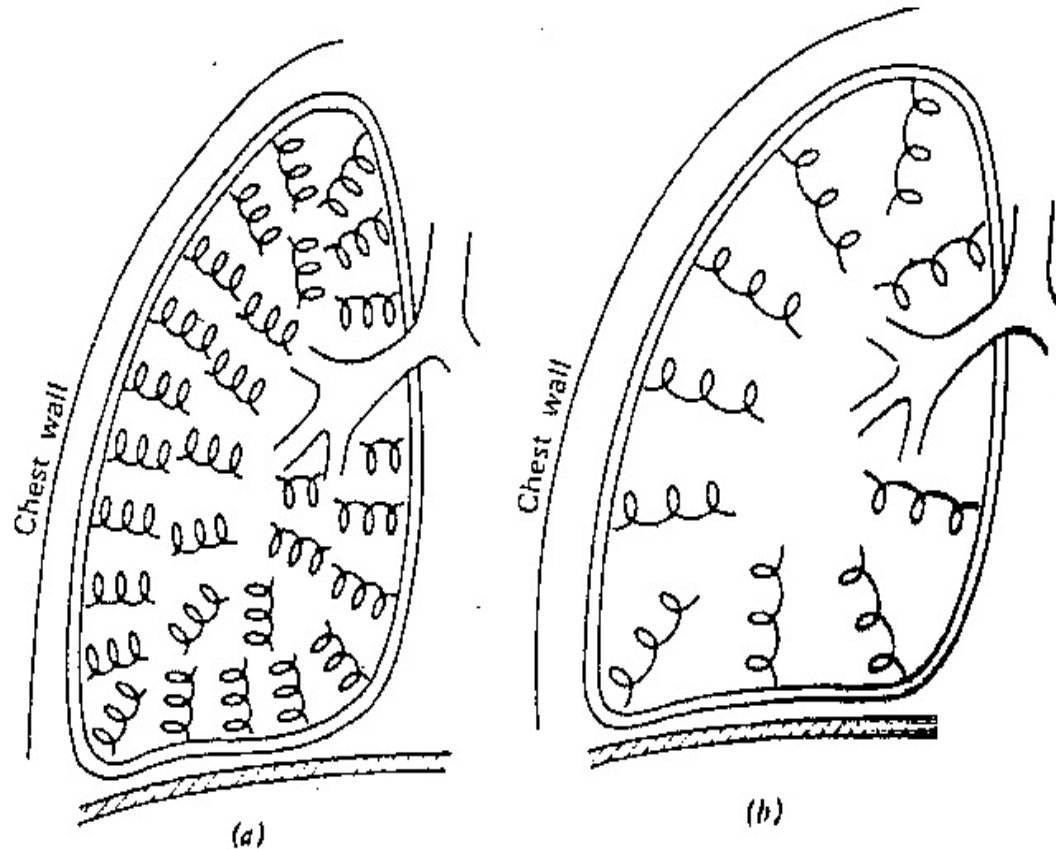
Physics of Some Common Lung Diseases

- a small change in pressure produces a larger than normal change in the volume
- Much of the work of breathing is done in overcoming the resistance of the airways
- the airway resistance increases greatly.

Physics of Some Common Lung Diseases



Physics of Some Common Lung Diseases



Physics of Some Common L Diseases

- ❑ This produces two important changes:
 - the lung becomes
 - flappy
 - expands as the reduced tension allows the chest wall to expand
 - almost to the resting volume of the chest wall without the lung- about 60% of vital capacity

Physics of Some Common Lung Diseases

- The tissues do not pull very hard on the airways,
 - permitting the narrowed airways to collapse easily during expiration.
- ❑ This increased airway resistance is the major symptom of sever emphysema.
- ❑ The increased size of the lungs increases the FRC and the residual volume

Physics of Some Common L Diseases

- ❑ The chest is overinflated,
- ❑ the posture is affected:
 - someone with the disease appear barrel-chested
 - who has emphysema is unable to blow out a candle,

Physics of Some Common Lung Diseases

- ❑ In asthma, obstructive
 - The expiratory difficulty due to increased airway resistance.
 - due to swelling (edema) and
 - mucus in the smaller airways,
 - contraction of the smooth muscle around the large airways.
 - Lung compliance is essentially normal,

Physics of Some Common L Diseases

- FRC may be higher than normal since the patient often starts to inspire before completing a normal expiration.

Physics of Some Common Lung Diseases

- ❑ In fibrosis of the lungs the membranes between the alveoli thicken.
- This has two marked effects:
 - The compliance of the lung decreases
 - The diffusion of O_2 into the pulmonary capillaries decreases

Physics of Some Common Lung Diseases

- The expiratory resistance is essentially normal.
- A person with the disease will have labored and even painful breathing (dyspnea)
- shortness of breath during exercise.
- Fibrosis of the lungs can occur if the lungs have been irradiated (eig., in the treatment of cancer),

Problem

- 1. In Morochocha, Peru, the atmospheric pressure is 59 kPa (447 mmHg). Find the pO_2 and pN_2 that people in this village inhale.
- 2. A person's lung volumes were measured and the following result were obtained: vital capacity: 5 liters; residual volume 1.0 liter; and expiratory reserve volume: 1.5 liters. Find the functional residual capacity (FRC) of this individual.

Problems

- 3. A person inspired maximally and then began breathing from an expandable bag containing 2 liters of 40% helium gas. After a few breaths, the helium concentration in the bag was 10%. What was this person's total lung capacity?

Prove that

- $T = CXR_g = \text{sec?}$